In [14]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Optional
import warnings
warnings.filterwarnings('ignore')
```

In [15]:
```
df = pd.read_csv(r"C:\Users\DELL\superstore_final_dataset (1).csv", encoding='la
```

In [16]:
```
df.columns = df.columns.str.strip().str.replace('\ufeff', '').str.replace(' ', '
print(df.columns)
```

```
Index(['row_id', 'order_id', 'order_date', 'ship_date', 'ship_mode',
       'customer_id', 'customer_name', 'segment', 'country', 'city', 'state',
       'postal_code', 'region', 'product_id', 'category', 'sub_category',
       'product_name', 'sales'],
      dtype='object')
```

In [20]:
```
df.head()
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 3955 entries, 0 to 9799
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   row_id         3955 non-null   int64
 1   order_id       3955 non-null   object
 2   order_date     3955 non-null   datetime64[ns]
 3   ship_date      3955 non-null   object
 4   ship_mode      3955 non-null   object
 5   customer_id    3955 non-null   object
 6   customer_name  3955 non-null   object
 7   segment        3955 non-null   object
 8   country        3955 non-null   object
 9   city           3955 non-null   object
 10  state          3955 non-null   object
 11  postal_code    3955 non-null   float64
 12  region         3955 non-null   object
 13  product_id     3955 non-null   object
 14  category       3955 non-null   object
 15  sub_category   3955 non-null   object
 16  product_name   3955 non-null   object
 17  sales          3955 non-null   float64
 18  year           3955 non-null   int32
 19  month          3955 non-null   int32
 20  month_name     3955 non-null   object
dtypes: datetime64[ns](1), float64(2), int32(2), int64(1), object(15)
memory usage: 648.9+ KB
```

Out[20]:

| | row_id | order_date | postal_code | sales | year | r |
|---|---|---|---|---|---|---|
| **count** | 3955.000000 | 3955 | 3955.000000 | 3955.000000 | 3955.000000 | 3955.0 |
| **mean** | 4868.681416 | 2017-03-14 18:03:54.841972224 | 55074.172946 | 218.898418 | 2016.728445 | 6.4 |
| **min** | 1.000000 | 2015-01-02 00:00:00 | 1040.000000 | 0.556000 | 2015.000000 | 1.0 |
| **25%** | 2465.500000 | 2016-04-05 00:00:00 | 22204.000000 | 17.472000 | 2016.000000 | 3.0 |
| **50%** | 4884.000000 | 2017-05-02 00:00:00 | 55407.000000 | 55.008000 | 2017.000000 | 6.0 |
| **75%** | 7319.000000 | 2018-03-07 00:00:00 | 90006.000000 | 199.925000 | 2018.000000 | 9.0 |
| **max** | 9800.000000 | 2018-12-11 00:00:00 | 99301.000000 | 17499.950000 | 2018.000000 | 12.0 |
| **std** | 2821.087831 | NaN | 32226.772397 | 560.799822 | 1.119373 | 3.4 |

In [18]:
```
# Convert date column
df['order_date'] = pd.to_datetime(df['order_date'], errors='coerce')

# Drop missing values
df = df.dropna()

# Remove duplicates
df = df.drop_duplicates()
```

In [19]:
```
df['year'] = df['order_date'].dt.year
df['month'] = df['order_date'].dt.month
df['month_name'] = df['order_date'].dt.strftime('%b')
```

In [22]:
```
total_revenue = df['sales'].sum()
avg_order_value = df['sales'].mean()

print("Total Revenue:", total_revenue)
print("Average Order Value:", avg_order_value)
```

```
Total Revenue: 865743.2436
Average Order Value: 218.89841810366624
```

In [23]:
```
df['profit'] = df['sales'] * 0.1
```

In [24]:
```
total_profit = df['profit'].sum()
profit_margin = (total_profit / df['sales'].sum()) * 100

print("Total Profit:", total_profit)
print("Profit Margin:", profit_margin)
```

```
Total Profit: 86574.32436
Profit Margin: 10.0
```

In [25]:
```
print(df.columns)
```

```
Index(['row_id', 'order_id', 'order_date', 'ship_date', 'ship_mode',
       'customer_id', 'customer_name', 'segment', 'country', 'city', 'state',
       'postal_code', 'region', 'product_id', 'category', 'sub_category',
       'product_name', 'sales', 'year', 'month', 'month_name', 'profit'],
      dtype='object')
```

In [27]:

```
# KPIs
total_revenue = df['sales'].sum()
avg_order_value = df['sales'].mean()

# Optional profit
df['profit'] = df['sales'] * 0.1

total_profit = df['profit'].sum()

print("Revenue:", total_revenue)
print("Avg Order Value:", avg_order_value)
print("Estimated Profit:", total_profit)
```

```
Revenue: 865743.2436
Avg Order Value: 218.89841810366624
Estimated Profit: 86574.32436
```

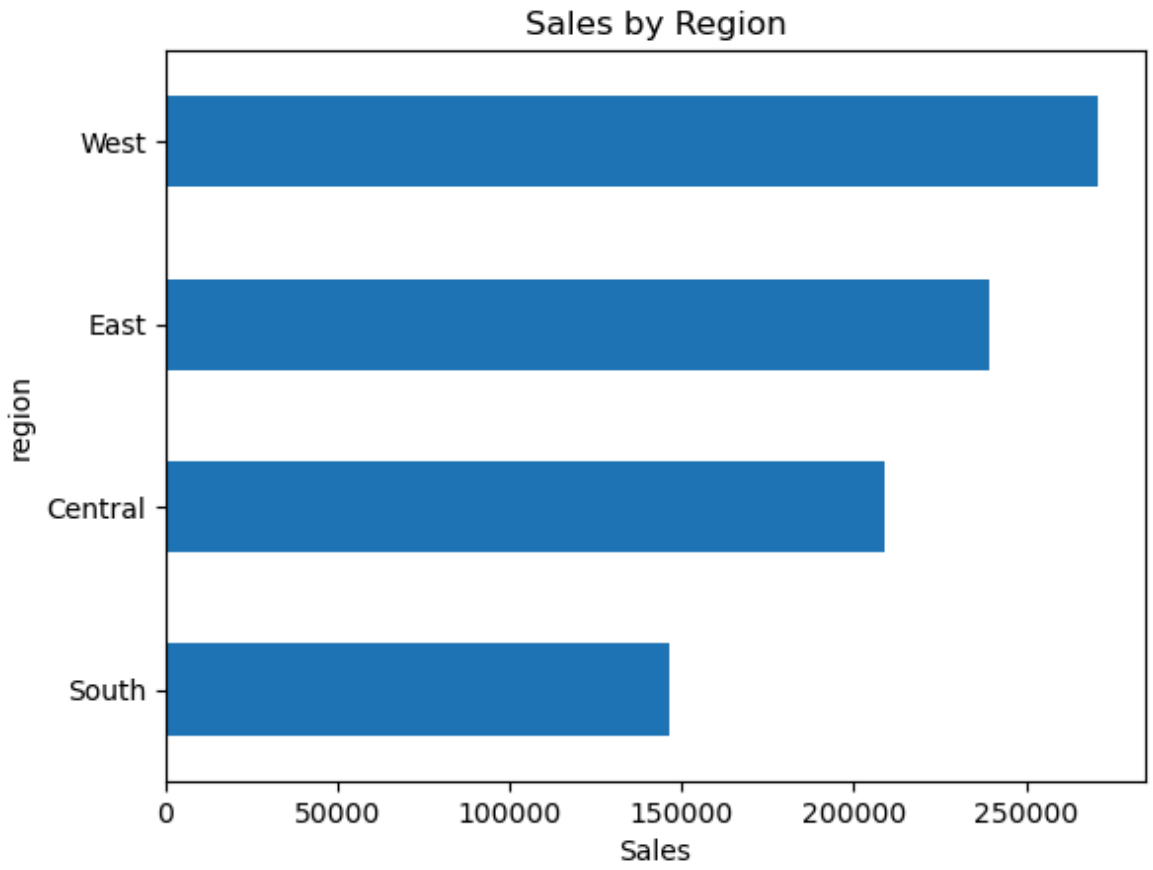
In [28]:

```
total_revenue = df['sales'].sum()
total_profit = df['profit'].sum()
profit_margin = (total_profit / total_revenue) * 100
avg_order_value = df['sales'].mean()

print("Total Revenue:", total_revenue)
print("Total Profit:", total_profit)
print("Profit Margin:", profit_margin)
print("Average Order Value:", avg_order_value)
```

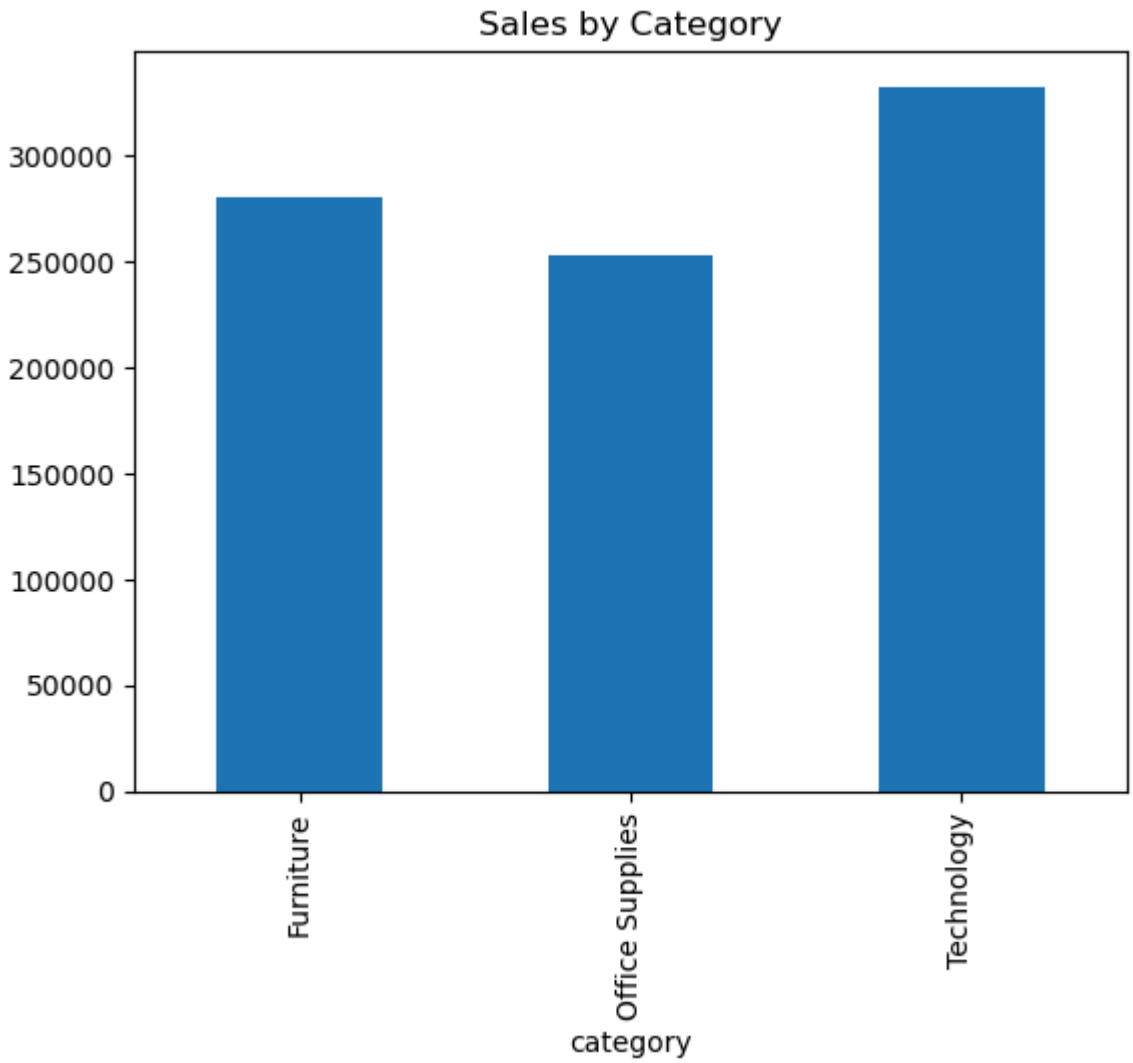
```
Total Revenue: 865743.2436
Total Profit: 86574.32436
Profit Margin: 10.0
Average Order Value: 218.89841810366624
```

In [29]:

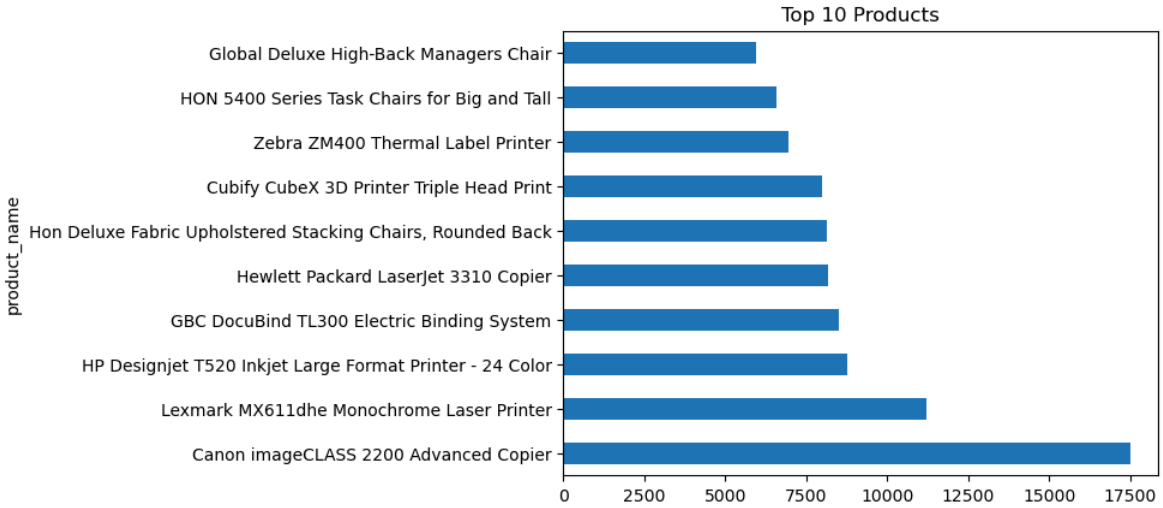
```
region_sales = df.groupby('region')['sales'].sum().sort_values()

region_sales.plot(kind='barh')
plt.title("Sales by Region")
plt.xlabel("Sales")
plt.show()
```

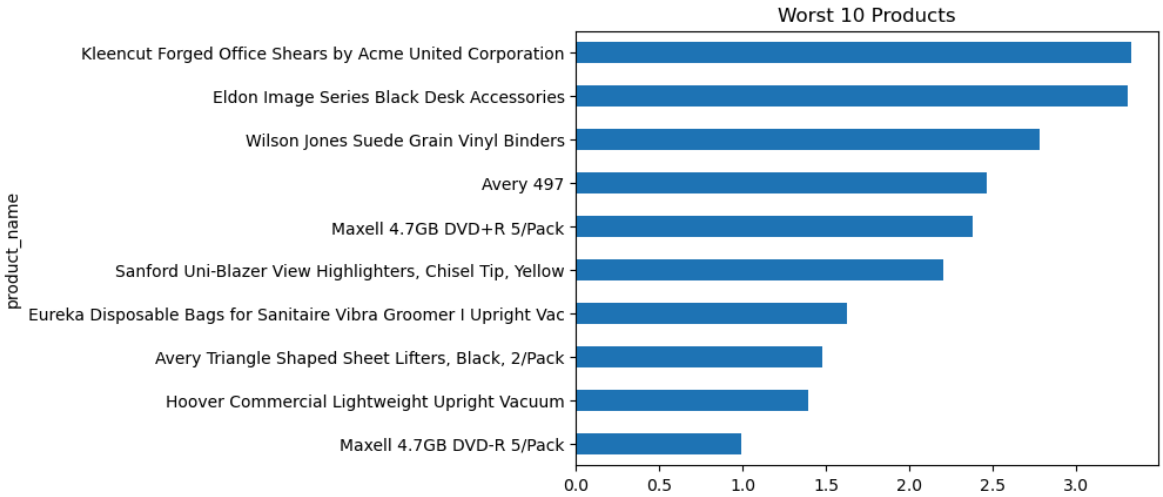



In [30]:
```
category_sales = df.groupby('category')['sales'].sum()

category_sales.plot(kind='bar')
plt.title("Sales by Category")
plt.show()
```

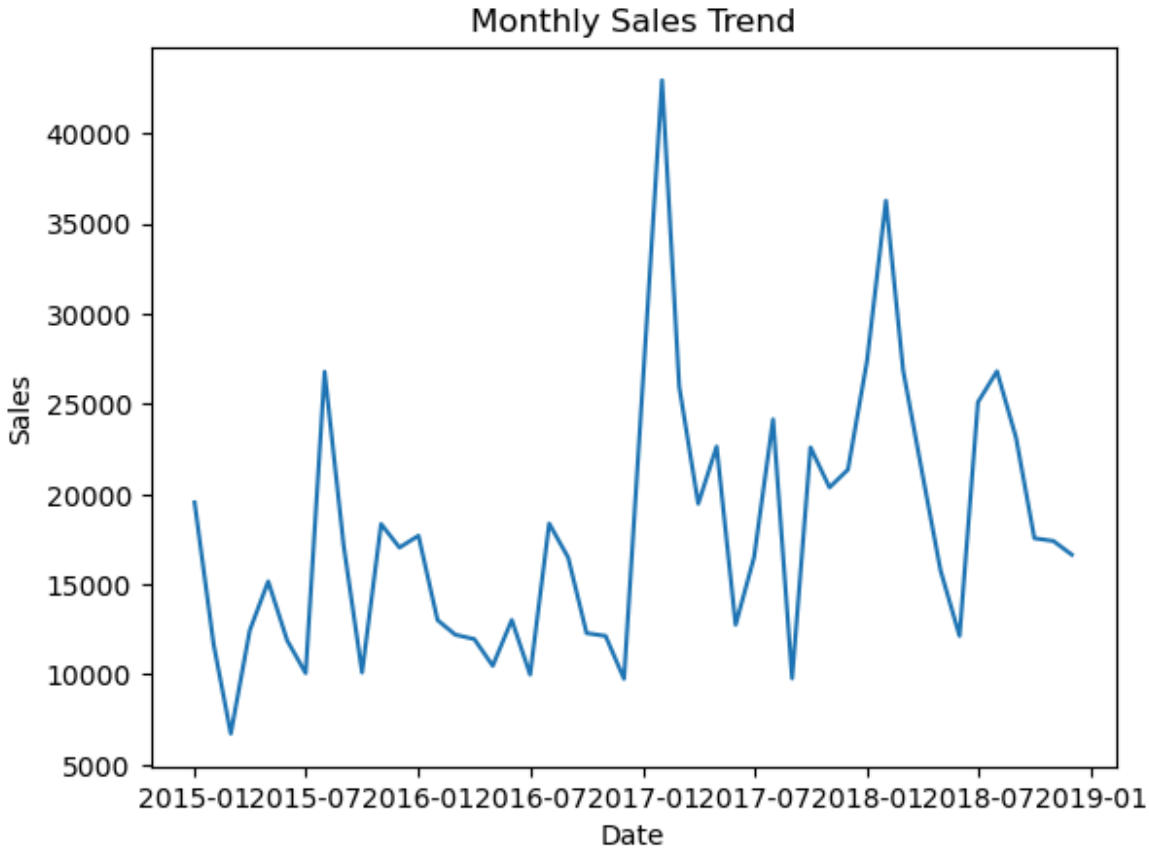



In [31]:
```
product_sales = df.groupby('product_name')['sales'].sum()

top_products = product_sales.sort_values(ascending=False).head(10)
worst_products = product_sales.sort_values().head(10)

# Top
top_products.plot(kind='barh')
plt.title("Top 10 Products")
plt.show()

# Worst
worst_products.plot(kind='barh')
plt.title("Worst 10 Products")
plt.show()
```

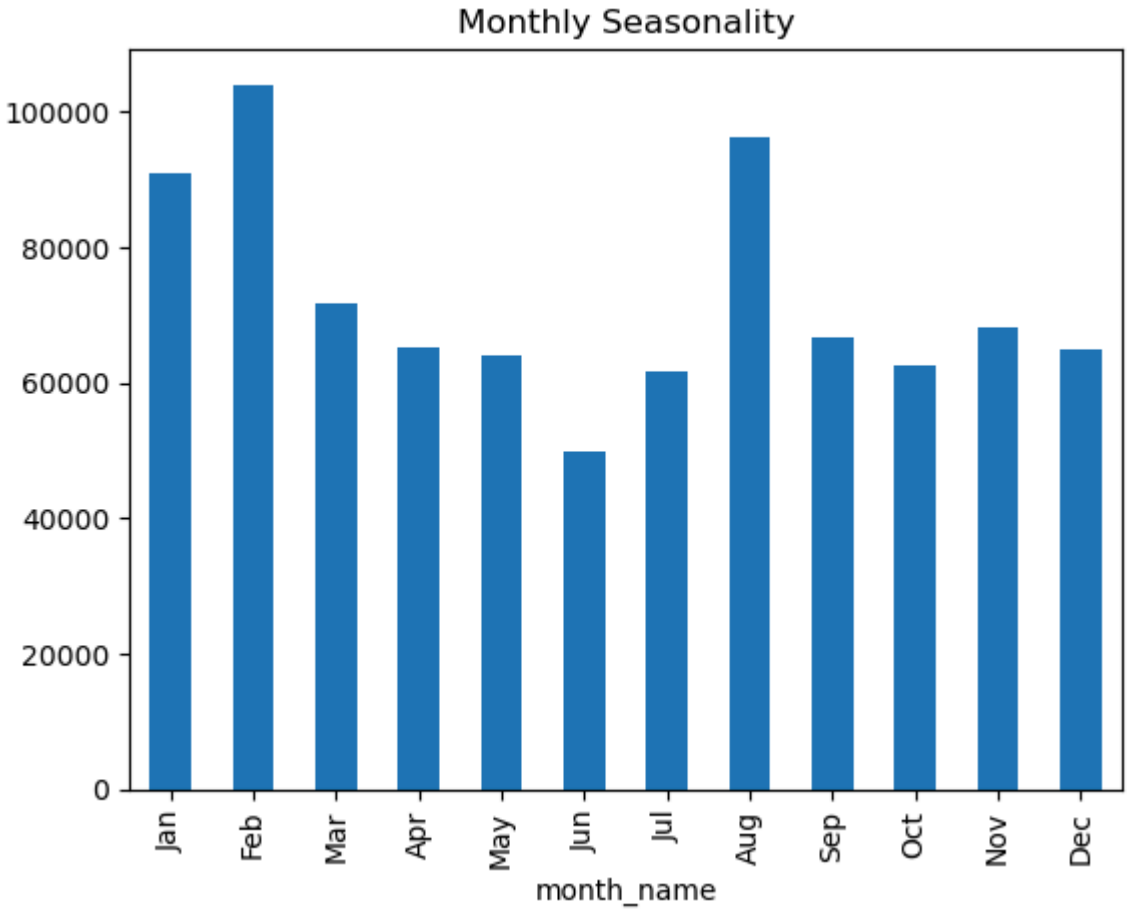


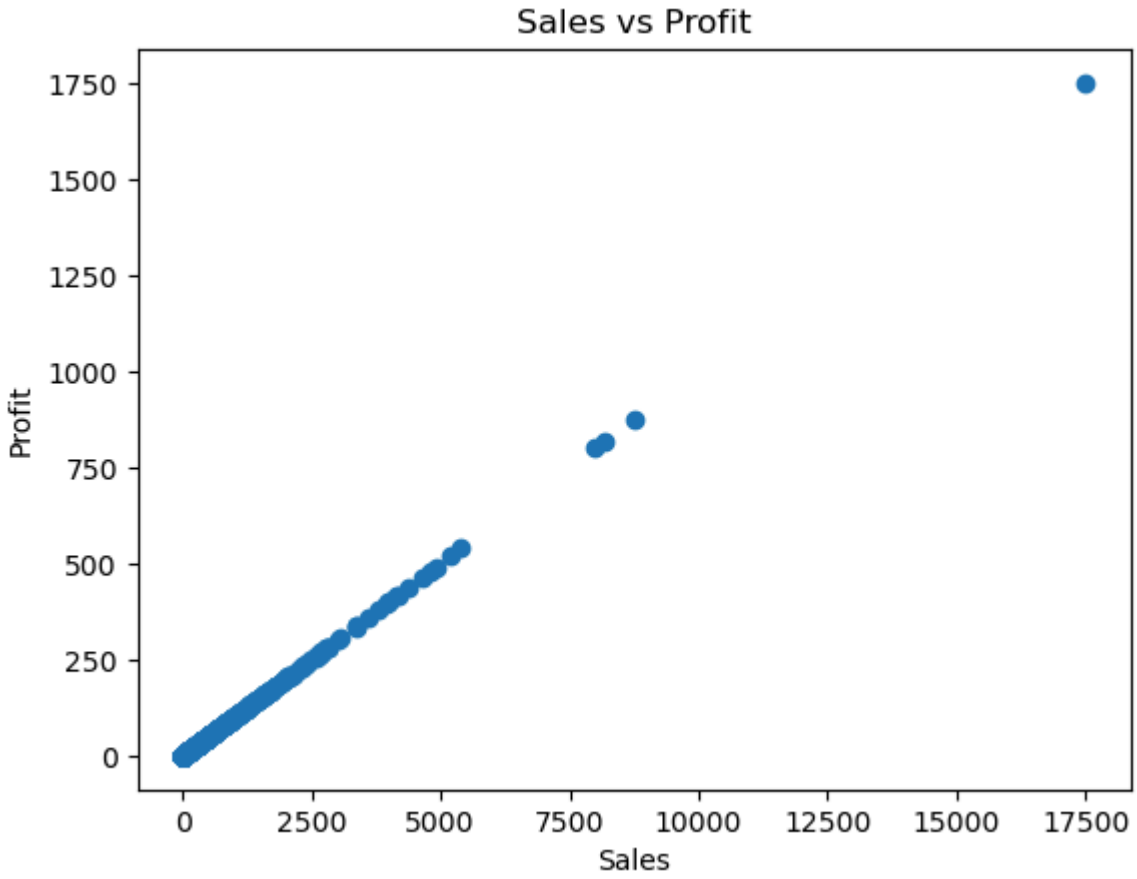

In [32]:
```
monthly_sales = df.groupby(['year','month'])['sales'].sum().reset_index()

monthly_sales['date'] = pd.to_datetime(monthly_sales[['year','month']].assign(da

plt.plot(monthly_sales['date'], monthly_sales['sales'])
plt.title("Monthly Sales Trend")
plt.xlabel("Date")
plt.ylabel("Sales")
plt.show()
```




In [33]:
```
monthwise = df.groupby('month_name')['sales'].sum()

monthwise = monthwise.reindex([
    'Jan','Feb','Mar','Apr','May','Jun',
    'Jul','Aug','Sep','Oct','Nov','Dec'
])

monthwise.plot(kind='bar')
plt.title("Monthly Seasonality")
plt.show()
```




In [34]:
```
plt.scatter(df['sales'], df['profit'])
plt.title("Sales vs Profit")
plt.xlabel("Sales")
plt.ylabel("Profit")
plt.show()
```




In [ ]:
```
Business Insights
Top Products generate majority revenue → Focus on them
Some products have high sales but low profit → Pricing issue
Certain months show low sales → Seasonality effect
Some regions underperform → Need targeted marketing
```

In [ ]:
```
Tactical Improvements
improvements = [
    "Focus marketing on high-performing regions",
    "Improve pricing strategy for low-profit products",
    "Offer discounts during low-sales months",
    "Bundle products to increase average order value",
    "Promote best-selling products more"
]

for i, imp in enumerate(improvements, 1):
    print(f"{i}. {imp}")
```

In [ ]: